

Bovnar (BVNR) — Units & Currencies Reference

Bovnar (BVNR) v1.1 documentation · Units & Currencies Cheat Sheet · 2026-06-19

Spec version 1.1 · 163 physical units · 164 fiat currencies · 50 cryptocurrencies

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6. Symbol Disambiguation

1. SI Prefixes

Written as `prefix~base` (mandatory `~` separator). Example: `k~m` = kilometre.

Name	Symbol	Factor	Enum (<code>si_prefix_id_t</code>)
quetta	<code>Q</code>	10^{30}	<code>si_quetta</code>
ronna	<code>R</code>	10^{27}	<code>si_ronna</code>
yotta	<code>Y</code>	10^{24}	<code>si_yotta</code>
zetta	<code>Z</code>	10^{21}	<code>si_zetta</code>
exa	<code>E</code>	10^{18}	<code>si_exa</code>

Name	Symbol	Factor	Enum (si_prefix_id_t)
peta	P	10 ¹⁵	si_peta
tera	T	10 ¹²	si_tera
giga	G	10 ⁹	si_giga
mega	M	10 ⁶	si_mega
kilo	k	10 ³	si_kilo
hecto	h	10 ²	si_hecto
deca	da	10 ¹	si_deca
(none)	—	10 ⁰	si_none
deci	d	10 ⁻¹	si_deci
centi	c	10 ⁻²	si_cent
milli	m	10 ⁻³	si_milli
micro	μ	10 ⁻⁶	si_micro
nano	n	10 ⁻⁹	si_nano
pico	p	10 ⁻¹²	si_pico
femto	f	10 ⁻¹⁵	si_femto
atto	a	10 ⁻¹⁸	si_atto
zepto	z	10 ⁻²¹	si_zepto
yocto	y	10 ⁻²⁴	si_yocto
ronto	r	10 ⁻²⁷	si_ronto
quecto	q	10 ⁻³⁰	si_quecto

μ is U+00B5 MICRO SIGN (UTF-8 0xC2 0xB5). U+03BC (Greek small letter mu) is **not** accepted. da is a two-character prefix: da~m = decametre.

Prefix-base ambiguities — resolved by the mandatory ~:

Bare token	Is a base unit	With ~ becomes prefix
m	meter (bu_meter)	milli
d	day (bu_day)	deci
h	hour (bu_hour)	hecto
T	tesla (bu_tesla)	tera
G	gauss (bu_gauss)	giga
P	poise (bu_poise)	peta
R	röntgen (bu_roentgen)	ronna

Bare token	Is a base unit	With ~ becomes prefix
f	farad (bu_farad)	femto
S	siemens (bu_siemens)	(not a prefix — S has no prefix role)

Examples: bare m = metre; m~s = millisecond. Bare d = day; d~s = decisecond.

2. IEC Binary Prefixes

Used **only** on b (bit) and B (byte). Written as prefix~base: Ki~B = kibibyte.

Name	Symbol	Factor	Enum (iec_prefix_id_t)
kibi	Ki	2 ¹⁰	iec_kibi
mebi	Mi	2 ²⁰	iec_mebi
gibi	Gi	2 ³⁰	iec_gibi
tebi	Ti	2 ⁴⁰	iec_tebi
pebi	Pi	2 ⁵⁰	iec_pebi
exbi	Ei	2 ⁶⁰	iec_exbi
zebi	Zi	2 ⁷⁰	iec_zebi
yobi	Yi	2 ⁸⁰	iec_yobi
robi	Ri	2 ⁹⁰	iec_robi
quebi	Qi	2 ¹⁰⁰	iec_quebi

3. Prefix Validity Rules

Unit category	SI prefixes	IEC prefixes
All physical units (default)	All 24 allowed	Forbidden
b (bit) and B (byte)	Only ≥ kilo (k , M , G , ..., Q)	All 10 allowed
b and B with sub-kilo SI (d , c , m , μ , n , p , f , a , z , y , r , q , da , h)	Forbidden	—
Currency codes	All 24 allowed	Forbidden
Old German units (bu_pfund ... bu_scheffel)	None (si_none only)	Forbidden

4. Physical Units

Symbol = canonical serialized form (produced on output; accepted on input). **Long forms** = accepted on input only; never produced on output. **Factor** = conversion factor to SI base units unless noted.

4.1 SI Base Units

Symbol	Long forms	Name	Enum
s	sec, second, seconds	second	bu_second
m	meter, metre, meters, metres	metre	bu_meter
g	gram, grams	gram	bu_gram
A	amp, amps, ampere, amperes	ampere	bu_ampere
K	kelvin, kelvins	kelvin	bu_kelvin
mol	mole, moles	mole	bu_mol
cd	candela, candelas	candela	bu_candela

g (gram) is the base symbol; k~g = kilogram.

4.2 Named SI-Derived Units

Symbol	Long forms	Name	Enum	SI definition
Hz	hertz	hertz	bu_hertz	s^{-1}
N	newton, newtons	newton	bu_newton	$kg \cdot m \cdot s^{-2}$
Pa	pascal, pascals	pascal	bu_pascal	$kg \cdot m^{-1} \cdot s^{-2}$
J	joule, joules	joule	bu_joule	$kg \cdot m^2 \cdot s^{-2}$
W	watt, watts	watt	bu_watt	$kg \cdot m^2 \cdot s^{-3}$
V	volt, volts	volt	bu_volt	$kg \cdot m^2 \cdot A^{-1} \cdot s^{-3}$
Ω	ohm, ohms, Ohm	ohm	bu_ohm	$kg \cdot m^2 \cdot A^{-2} \cdot s^{-3}$ — U+2126 OHM SIGN; U+03A9 (Greek capital omega) not accepted
F	farad, farads	farad	bu_farad	$kg^{-1} \cdot m^{-2} \cdot A^2 \cdot s^4$
C	coulomb, coulombs	coulomb	bu_coulomb	$A \cdot s$
S	siemens	siemens	bu_siemens	$kg^{-1} \cdot m^{-2} \cdot A^2 \cdot s^3$
Wb	weber, webers	weber	bu_weber	$kg \cdot m^2 \cdot A^{-1} \cdot s^{-2}$

Symbol	Long forms	Name	Enum	SI definition
T	tesla, teslas	tesla	bu_tesla	$\text{kg}\cdot\text{A}^{-1}\cdot\text{s}^{-2}$
H	henry, henrys, henries	henry	bu_henry	$\text{kg}\cdot\text{m}^2\cdot\text{A}^{-2}\cdot\text{s}^{-2}$
lm	lumen, lumens	lumen	bu_lumen	$\text{cd}\cdot\text{s}$
lx	lux	lux	bu_lux	$\text{cd}\cdot\text{s}\cdot\text{m}^{-2}$
Bq	becquerel, becquerels	becquerel	bu_becquerel	s^{-1}
Gy	gray, grays	gray	bu_gray	$\text{m}^2\cdot\text{s}^{-2}$
Sv	sievert, sieverts	sievert	bu_sievert	$\text{m}^2\cdot\text{s}^{-2}$
kat	katal, katalas	katal	bu_katal	$\text{mol}\cdot\text{s}^{-1}$
rad	radian, radians	radian	bu_radian	dimensionless (m/m)
sr	steradian, steradians	steradian	bu_steradian	dimensionless (m^2/m^2)

4.3 Non-SI Units Accepted with SI

Symbol	Long forms	Name	Enum	Factor / notes
L, l	liter, litre, liters, litres	litre	bu_liter	10^{-3} m^3
min	minute, minutes	minute	bu_minute	60 s
h	hour, hours	hour	bu_hour	3600 s
d	day, days	day	bu_day	86400 s
wk	week, weeks	week	bu_week	604 800 s
mo	month, months	month (Julian)	bu_month	2 629 800 s (= 365.25 d / 12)
fn	fortnight, fortnights	fortnight	bu_fortnight	1 209 600 s (= 14 d)
yr	year, years	year (Julian)	bu_year	31 557 600 s
°, deg	degr, degree, degrees	degree (angle)	bu_degree	$\pi/180 \text{ rad}$ — U+00B0
t	tonne	tonne	bu_tonne	10^3 kg
bar	—	bar	bu_bar	10^5 Pa
eV	electronvolt	electronvolt	bu_electronvolt	$1.602176634\times 10^{-19} \text{ J}$
Da	dalton	dalton	bu_dalton	$1.66053906660\times 10^{-27} \text{ kg}$

Symbol	Long forms	Name	Enum	Factor / notes
<code>au</code>	—	astronomical unit	<code>bu_astronomical_unit</code>	$1.495978707 \times 10^{11}$ m
<code>ha</code>	<code>hectare</code>	hectare	<code>bu_hectare</code>	10^4 m ²

4.4 Imperial & US Customary — Length

Symbol	Long forms	Name	Enum	Factor
<code>in</code>	<code>inch</code> , <code>inches</code>	inch	<code>bu_inch</code>	0.0254 m (exact)
<code>ft</code>	<code>foot</code> , <code>feet</code>	foot	<code>bu_foot</code>	0.3048 m (exact)
<code>yd</code>	<code>yard</code> , <code>yards</code>	yard	<code>bu_yard</code>	0.9144 m (exact)
<code>mi</code>	<code>mile</code> , <code>miles</code>	statute mile	<code>bu_mile</code>	1609.344 m (exact)
<code>nmi</code>	<code>nautical_mile</code> , <code>nautical_miles</code>	nautical mile	<code>bu_nautical_mile</code>	1852 m (exact)
<code>Å</code> (U+212B)	<code>angstrom</code> , <code>angstroms</code> , <code>Å</code> (U+00C5)	ångström	<code>bu_angstrom</code>	10^{-10} m
<code>ly</code>	<code>light_year</code> , <code>light_years</code>	light-year	<code>bu_light_year</code>	$9.4607304725808 \times 10^{15}$ m
<code>pc</code>	<code>parsec</code> , <code>parsecs</code>	parsec	<code>bu_parsec</code>	$3.085677581491367 \times 10^{16}$ m
<code>fur</code>	<code>furlong</code> , <code>furlongs</code>	furlong	<code>bu_furlong</code>	201.168 m (exact)
<code>fath</code>	<code>fathom</code> , <code>fathoms</code>	fathom	<code>bu_fathom</code>	1.8288 m (exact)
<code>thou</code>	<code>thou</code> , <code>mil</code> , <code>mils</code>	thou	<code>bu_thou</code>	25.4×10^{-6} m (exact)
<code>ch</code>	<code>chain</code> , <code>chains</code>	chain (Gunter's)	<code>bu_chain</code>	20.1168 m (exact)
<code>rd</code>	<code>rod</code> , <code>rods</code>	rod (pole, perch)	<code>bu_rod</code>	5.0292 m (exact)

`thou` and `mil` are synonyms. Canonical output is `thou`. `mil` does **not** mean milliradian; milliradians are written `m~rad`.

4.5 Imperial & US Customary — Mass

Symbol	Long forms	Name	Enum	Factor
<code>lb</code>	<code>lbs</code> , <code>pound</code> , <code>pounds</code>	pound (avoirdupois)	<code>bu_pound</code>	0.45359237 kg (exact)
<code>oz</code>	<code>ounce</code> , <code>ounces</code>	ounce (avoirdupois)	<code>bu_ounce</code>	0.028349523125 kg (exact)

Symbol	Long forms	Name	Enum	Factor
<code>gr</code>	<code>grain</code> , <code>grains</code>	grain	<code>bu_grain</code>	6.479891×10 ⁻⁵ kg (exact)
<code>st</code>	<code>stone</code> , <code>stones</code>	stone	<code>bu_stone</code>	6.35029318 kg (exact)
<code>tn_sh</code>	<code>short_ton</code> , <code>short_tons</code>	short ton (US)	<code>bu_short_ton</code>	907.18474 kg (exact)
<code>tn_l</code>	<code>long_ton</code> , <code>long_tons</code>	long ton (UK)	<code>bu_long_ton</code>	1016.0469088 kg (exact)
<code>oz_t</code>	<code>troy_ounce</code> , <code>troy_ounces</code>	troy ounce	<code>bu_troy_ounce</code>	0.0311034768 kg (exact)
<code>ct</code>	<code>carat</code> , <code>carats</code>	metric carat	<code>bu_carat</code>	2×10 ⁻⁴ kg (exact)
<code>slug</code>	<code>slugs</code>	slug	<code>bu_slug</code>	14.593902937 kg
<code>dr</code>	<code>dram</code> , <code>drams</code>	dram (avoirdupois)	<code>bu_dram</code>	1.7718451953125×10 ⁻³ kg (exact)
<code>dwt</code>	<code>pennyweight</code> , <code>pennyweights</code>	pennyweight (troy)	<code>bu_pennyweight</code>	1.55517384×10 ⁻³ kg (exact)

4.6 Temperature

Symbol	Long forms	Name	Enum	Conversion
<code>°C</code> , <code>degC</code>	<code>degrC</code> , <code>degreeC</code> , <code>degreesC</code> , <code>celsius</code>	degree Celsius	<code>bu_celsius</code>	$K = ^\circ C + 273.15$ (affine)
<code>°F</code> , <code>degF</code>	<code>degrF</code> , <code>degreeF</code> , <code>degreesF</code> , <code>fahrenheit</code>	degree Fahrenheit	<code>bu_fahrenheit</code>	$K = (^\circ F + 459.67) \times 5/9$ (affine)
<code>°Ra</code> , <code>degRa</code>	<code>degrRa</code> , <code>degreeRa</code> , <code>degreesRa</code> , <code>rankine</code>	degree Rankine	<code>bu_rankine</code>	$K = ^\circ Ra \times 5/9$ (linear)
<code>°De</code> , <code>degDe</code>	<code>degrDe</code> , <code>degreeDe</code> , <code>degreesDe</code> , <code>delisle</code>	degree Delisle	<code>bu_delisle</code>	$K = 373.15 - ^\circ De \times 2/3$ (affine)
<code>°N</code> , <code>degN</code>	<code>degrN</code> , <code>degreeN</code> , <code>degreesN</code> , <code>newton_temperature</code>	degree Newton	<code>bu_newton_temp</code>	$K = ^\circ N \times 100/33 + 273.15$ (affine)
<code>°Re</code> , <code>degRe</code>	<code>degrRe</code> , <code>degreeRe</code> , <code>degreesRe</code> , <code>reaumur</code>	degree Réaumur	<code>bu_reaumur</code>	$K = ^\circ Re \times 5/4 + 273.15$ (affine)
<code>°Ro</code> , <code>degRo</code>	<code>degrRo</code> , <code>degreeRo</code> , <code>degreesRo</code> , <code>romer</code>	degree Rømer	<code>bu_romer</code>	$K = (^\circ Ro - 7.5) \times 40/21 + 273.15$ (affine)

Kelvin (**K**) is in §4.1. **R** is reserved for röntgen (§4.20). **N** alone is newton (§4.2); use `°N` or `degN` for Newton temperature.

4.7 Pressure

Symbol	Long forms	Name	Enum	Factor
<code>atm</code>	<code>atmosphere</code> , <code>atmospheres</code>	standard atmosphere	<code>bu_atmosphere</code>	101 325 Pa (exact)
<code>at</code>	<code>atmosphere_technical</code>	atmosphere technical	<code>bu_atmosphere_technical</code>	98 066.5 Pa (= 1 kgf/cm ²)
<code>mmHg</code>	—	millimetre of mercury	<code>bu_mmhg</code>	133.322387415 Pa
<code>Torr</code>	<code>torr</code>	torr	<code>bu_torr</code>	101 325/760 Pa
<code>psi</code>	—	pound-force per square inch	<code>bu_psi</code>	6894.757293168361 Pa
<code>inHg</code>	<code>inch_hg</code> , <code>inch_mercury</code>	inch of mercury	<code>bu_inch_hg</code>	3386.388645 Pa

`at` ≠ `atm`: 1 `at` = 98 066.5 Pa; 1 `atm` = 101 325 Pa.

4.8 Energy

Symbol	Long forms	Name	Enum	Factor
<code>cal</code>	<code>calorie</code> , <code>calories</code>	thermochemical calorie	<code>bu_calorie</code>	4.184 J (exact)
<code>Btu</code>	<code>btu</code> , <code>BTU</code>	International Table BTU	<code>bu_btu</code>	1055.05585262 J
<code>erg</code>	<code>ergs</code>	erg	<code>bu_erg</code>	10 ⁻⁷ J (exact)
<code>thm</code>	<code>therm</code> , <code>therms</code>	US therm	<code>bu_therm</code>	1.05480400×10 ⁸ J (exact)
<code>ft_lb</code>	<code>foot_pound</code> , <code>foot_pounds</code>	foot-pound	<code>bu_foot_pound</code>	1.3558179483 J

`BTU` (all-caps) is a valid alias: with no `$` sigil the token is a physical-unit lookup (the currency table is only consulted for `$`-prefixed tokens), and it matches `bu_btu`. `Btu` and `btu` are also accepted.

4.9 Power

Symbol	Long forms	Name	Enum	Factor
<code>hp</code>	<code>horsepower</code>	mechanical horsepower	<code>bu_horsepower</code>	745.69987158227 W

Symbol	Long forms	Name	Enum	Factor
PS	CV, metric_horsepower	metric horsepower	bu_metric_horsepower	735.49875 W (exact)

4.10 Force

Symbol	Long forms	Name	Enum	Factor
lbf	pound_force	pound-force	bu_pound_force	4.4482216152605 N
dyn	dyne, dynes	dyne	bu_dyne	10 ⁻⁵ N (exact)
kip	kips	kip (kilopound-force)	bu_kip	4448.2216152605 N
kgf	kilogram_force	kilogram-force	bu_kilogram_force	9.80665 N (exact)

4.11 Speed & Rotation

Symbol	Long forms	Name	Enum	Factor
kn	knot, knots	knot	bu_knot	1852/3600 m/s
rpm	—	revolutions per minute	bu_rpm	1/60 s ⁻¹

4.12 Acceleration

Symbol	Long forms	Name	Enum	Factor
gn	standard_gravity	standard gravity	bu_standard_gravity	9.80665 m·s ⁻² (exact)

4.13 Volume — US Liquid

Symbol	Long forms	Name	Enum	Factor
gal	gallon, gallons	US liquid gallon	bu_gallon	3.785411784×10 ⁻³ m ³ (exact)
qt	quart, quarts	US liquid quart	bu_quart	9.46352946×10 ⁻⁴ m ³
pt	pint, pints	US liquid pint	bu_pint	4.73176473×10 ⁻⁴ m ³
cup	cups	US cup	bu_cup	2.365882365×10 ⁻⁴ m ³
gi	gill, gills	US gill	bu_gill	1.18294118250×10 ⁻⁴ m ³
fl_oz	fluid_ounce, fluid_ounces	US fluid ounce	bu_fluid_ounce	2.95735295625×10 ⁻⁵ m ³
tbsp	tablespoon, tablespoons	US tablespoon	bu_tablespoon	1.47867648×10 ⁻⁵ m ³
tsp	teaspoon, teaspoons	US teaspoon	bu_teaspoon	4.92892159375×10 ⁻⁶ m ³
bbl	barrel, barrels	petroleum barrel	bu_barrel	0.158987294928 m ³

`cup` (lowercase) = US cup; `CUP` (uppercase) = Cuban Peso. See §6.

4.14 Volume — UK Imperial

Symbol	Long forms	Name	Enum	Factor
<code>gal_uk</code>	<code>gallon_uk</code> , <code>gallons_uk</code>	imperial gallon	<code>bu_gallon_uk</code>	$4.54609 \times 10^{-3} \text{ m}^3$ (exact)
<code>qt_uk</code>	<code>quart_uk</code> , <code>quarts_uk</code>	imperial quart	<code>bu_quart_uk</code>	$1136.5225 \times 10^{-6} \text{ m}^3$
<code>pt_uk</code>	<code>pint_uk</code> , <code>pints_uk</code>	imperial pint	<code>bu_pint_uk</code>	$568.26125 \times 10^{-6} \text{ m}^3$
<code>gi_uk</code>	<code>gill_uk</code> , <code>gills_uk</code>	imperial gill	<code>bu_gill_uk</code>	$1.420653125 \times 10^{-4} \text{ m}^3$ (exact)
<code>fl_oz_uk</code>	<code>fluid_ounce_uk</code> , <code>fluid_ounces_uk</code>	imperial fluid ounce	<code>bu_fluid_ounce_uk</code>	$28.4130625 \times 10^{-6} \text{ m}^3$

4.15 Volume — US Apothecary & Dry

Symbol	Long forms	Name	Enum	Factor
<code>fl_dr</code>	<code>fluid_dram</code> , <code>fluid_drams</code>	US fluid dram	<code>bu_fluid_dram</code>	$3.6966911953125 \times 10^{-6} \text{ m}^3$
<code>minim</code>	<code>minims</code>	US minim	<code>bu_minim</code>	$6.16115199218750 \times 10^{-8} \text{ m}^3$
<code>pk</code>	<code>peck</code> , <code>pecks</code>	US dry peck	<code>bu_peck</code>	$8.80976754172 \times 10^{-3} \text{ m}^3$
<code>bsh</code>	<code>bushel</code> , <code>bushels</code>	US bushel	<code>bu_bushel</code>	$3.523907016688 \times 10^{-2} \text{ m}^3$

`minim` not `min` (which is the minute).

4.16 Area

Symbol	Long forms	Name	Enum	Factor
<code>ac</code>	<code>acre</code> , <code>acres</code>	acre	<code>bu_acre</code>	4046.8564224 m^2 (exact)
<code>barn</code>	<code>barns</code>	barn	<code>bu_barn</code>	10^{-28} m^2 (exact)

4.17 Angle

Symbol	Long forms	Name	Enum	Factor
<code>arcmin</code>	<code>arcminute</code> , <code>arcminutes</code>	arcminute	<code>bu_arcminute</code>	$\pi/10800 \text{ rad}$
<code>arcsec</code>	<code>arcsecond</code> , <code>arcseconds</code>	arcsecond	<code>bu_arcsecond</code>	$\pi/648000 \text{ rad}$

Symbol	Long forms	Name	Enum	Factor
grad	gradian, gradians, gon	gradian	bu_grad	$\pi/200$ rad
rev	turn, revolution, revolutions, turns	revolution	bu_revolution	2π rad

4.18 Digital

Symbol	Long forms	Name	Enum
b	bit, bits	bit	bu_bit
B	byte, bytes, Byte, Bytes	byte	bu_byte

4.19 CGS Units

Symbol	Long forms	Name	Enum	SI equivalent
P	poise, poises	poise (dynamic viscosity)	bu_poise	0.1 Pa·s
St	stokes, stoke	stokes (kinematic viscosity)	bu_stokes	10^{-4} m ² ·s ⁻¹
G	gauss	gauss (magnetic flux density)	bu_gauss	10^{-4} T
Mx	maxwell, maxwells	maxwell (magnetic flux)	bu_maxwell	10^{-8} Wb
Oe	oersted, oersteds	oersted (magnetic field strength)	bu_oersted	$1000/(4\pi)$ A/m
sb	stilb, stilbs	stilb (luminance)	bu_stilb	10^4 cd/m ²
ph	phot, phots	phot (illuminance)	bu_phot	10^4 lx
Gal	galileo, galileos	galileo (acceleration)	bu_galileo	10^{-2} m/s ²

4.20 Radiation

Symbol	Long forms	Name	Enum	SI equivalent
Ci	curie, curies	curie (radioactivity)	bu_curie	3.7×10^{10} Bq
R	roentgen, roentgens	röntgen (radiation exposure)	bu_roentgen	2.58×10^{-4} C/kg
rem	rems	rem (dose equivalent)	bu_rem	10^{-2} Sv

4.21 Logarithmic

Symbol	Long forms	Name	Enum	Notes
Np	neper, nepers	neper	bu_neper	dimensionless; $1 \text{ Np} = 20/\ln(10) \text{ dB} \approx 8.686 \text{ dB}$
dB	decibel, decibels	decibel	bu_decibel	dimensionless

4.22 Electrical Power

Symbol	Long forms	Name	Enum	Notes
<code>var</code>	<code>vars</code>	var (volt-ampere reactive)	<code>bu_var</code>	reactive power; same SI dim as W
<code>VA</code>	<code>volt_ampere</code> , <code>volt_amperes</code>	volt-ampere	<code>bu_volt_ampere</code>	apparent power; same SI dim as W

`W`, `var`, and `VA` all have SI dimension $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}$. `bvn_units_compatible` returns `true` across them; use `.components[0].base` to distinguish.

4.23 Textile Linear Density

Symbol	Long forms	Name	Enum	Factor
<code>tex</code>	—	tex	<code>bu_tex</code>	1×10^{-6} kg/m (ISO 1144)
<code>den</code>	<code>denier</code> , <code>deniers</code>	denier	<code>bu_denier</code>	1/9 000 000 kg/m

9 den = 1 tex.

4.24 Old German Units

No Old German unit accepts any SI or IEC prefix (`bvn_prefix_unit_valid` rejects all non-`si_none` prefixes for `bu_pfund` ... `bu_scheffel`). Enum values 348–360.

Mass (metric-compatible)

Symbol	Long forms	Name	Enum	Factor
<code>Pfd</code>	<code>pfund</code> , <code>pfunds</code>	Pfund	<code>bu_pfund</code>	0.5 kg (exact)
<code>Ztr</code>	<code>zentner</code>	Zentner	<code>bu_zentner</code>	50 kg (exact)
<code>dz</code>	<code>doppelzentner</code>	Doppelzentner	<code>bu_doppelzentner</code>	100 kg (exact)
<code>lot</code>	<code>lots</code>	Lot	<code>bu_lot</code>	15.625×10^{-3} kg (exact)

Length (historical Prussian)

Symbol	Long forms	Name	Enum	Factor
<code>prln</code>	<code>prussian_line</code> , <code>linie</code>	Prussian line	<code>bu_prussian_line</code>	2.17953×10^{-3} m
<code>prz</code>	<code>prussian_zoll</code> , <code>zoll</code>	Prussian Zoll	<code>bu_prussian_zoll</code>	2.61544×10^{-2} m
<code>prf</code>	<code>prussian_fuss</code> , <code>preussischer_fuss</code>	Prussian Fuß	<code>bu_prussian_fuss</code>	3.13853×10^{-1} m

Symbol	Long forms	Name	Enum	Factor
elle	prussian_elle, preussische_elle	Prussian Elle	bu_prussian_elle	6.67160×10 ⁻¹ m
rute	prussian_rute, preussische_rute	Prussian Rute	bu_prussian_rute	3.76624 m
klafter	prussian_klafter	Klafter	bu_klafter	1.88312 m
dt_mi	deutsche_meile, german_mile	Geographische Meile	bu_german_mile	7420.44 m

Area (historical Prussian)

Symbol	Long forms	Name	Enum	Factor
morgen	prussian_morgen	Morgen (Prussian)	bu_morgen	2553.22 m ²

Volume (historical Prussian)

Symbol	Long forms	Name	Enum	Factor
schffl	scheffel, prussian_scheffel	Scheffel (Prussian)	bu_scheffel	54.961×10 ⁻³ m ³

4.25 Additional Physical Units (361-367)

Length

Symbol	Long forms	Name	Enum	Factor
ftUS	survey_foot	US survey foot	bu_survey_foot	1200/3937 m ≈ 0.30480061 m
lea	league, leagues	League (US statute, 3 mi)	bu_league	4828.032 m
cbl	cable, cables	Cable (international, 1/10 nmi)	bu_cable	185.2 m
hand	hands	Hand (4 in)	bu_hand	0.1016 m

Mass

Symbol	Long forms	Name	Enum	Factor
qntl	quintal, quintals	Metric quintal	bu_quintal	100 kg
sc	scruple, scruples	Apothecary scruple (20 gr)	bu_scruple	1.2959782×10 ⁻³ kg

Signal Rate

Symbol	Long forms	Name	Enum	SI dimension
Bd	baud, bauds	Baud (symbol/s)	bu_baud	s ⁻¹

SI prefixes are accepted on all units in this section. IEC prefixes are rejected for all non-digital units.

4.26 Ratio and Proportion (372-377)

Dimensionless scaling factors: `5 %` $\equiv$ `0.05`, `250 ppm` $\equiv$ `0.00025`. These do **not** accept SI or IEC prefixes.

Symbol	Long forms	Name	Enum	Factor
<code>%</code>	<code>percent</code>	per cent	<code>bu_percent</code>	10^{-2}
<code>‰</code>	<code>per_mille</code>	per mille	<code>bu_per_mille</code>	10^{-3}
<code>‱</code>	<code>per_myriad</code>	per myriad	<code>bu_per_myriad</code>	10^{-4}
<code>pcm</code>	<code>per_cent_mille</code>	per cent mille	<code>bu_per_cent_mille</code>	10^{-5}
<code>ppm</code>	—	parts per million	<code>bu_ppm</code>	10^{-6}
<code>ppb</code>	—	parts per billion	<code>bu_ppb</code>	10^{-9}

5. Currencies

5.1 The Mandatory Currency Sigil

As of spec 1.0 a currency code carries a **mandatory** `$` **sigil**: write `$USD`, `$BTC`, `k~$EUR`, `$USD/oz_t`. The codes listed in the tables below are the bare ISO 4217 / crypto identifiers — prefix each with `$` when you use it in a document. A bare code (no `$`) is **not** a currency; it is matched against the physical-unit table and raises `error_unit_illegal` if it is not a unit. This removes every currency/unit namespace collision (e.g. `$CUP` the Cuban Peso vs `cup` the unit).

5.2 ISO 4217 Fiat Currencies

164 codes occupying `value_base_unit_t` slots **134 ... 297** (`BVN_CURRENCY_FIAT_FIRST ... BVN_CURRENCY_FIAT_LAST`). Currencies have no named `bu_*` enumerators — they are resolved from the `$`-sigil code by `bnv_parse_currency_str` and carried as the numeric `base` value; query them with `bnv_unit_is_fiat` / `bnv_currency_info`.

Min = minor unit exponent N: 1 major unit = 10^N minor units (e.g. 1 USD = 100 cents, N=2). Minor units are **bold** when they differ from 2. `numeric_code` is the ISO 4217 numeric identifier.

Code	Num	Min	Name
<code>AED</code>	784	2	UAE Dirham

Code	Num	Min	Name
AFN	971	2	Afghan Afghani
ALL	8	2	Albanian Lek
AMD	51	2	Armenian Dram
ANG	532	2	Netherlands Antillean Guilder
AOA	973	2	Angolan Kwanza
ARS	32	2	Argentine Peso
AUD	36	2	Australian Dollar
AWG	533	2	Aruban Florin
AZN	944	2	Azerbaijani Manat
BAM	977	2	Bosnia-Herzegovina Convertible Mark
BBD	52	2	Barbados Dollar
BDT	50	2	Bangladeshi Taka
BGN	975	2	Bulgarian Lev
BHD	48	3	Bahraini Dinar
BIF	108	0	Burundian Franc
BMD	60	2	Bermudian Dollar
BND	96	2	Brunei Dollar
BOB	68	2	Boliviano
BRL	986	2	Brazilian Real
BSD	44	2	Bahamian Dollar
BTN	64	2	Bhutanese Ngultrum
BWP	72	2	Botswana Pula
BYN	933	2	Belarusian Ruble
BZD	84	2	Belize Dollar
CAD	124	2	Canadian Dollar
CDF	976	2	Congolese Franc
CHF	756	2	Swiss Franc
CLF	990	4	Unidad de Fomento
CLP	152	0	Chilean Peso
CNY	156	2	Chinese Yuan
COP	170	2	Colombian Peso
CRC	188	2	Costa Rican Colon

Code	Num	Min	Name
CUP	192	2	Cuban Peso
CVE	132	2	Cape Verdean Escudo
CZK	203	2	Czech Koruna
DJF	262	0	Djiboutian Franc
DKK	208	2	Danish Krone
DOP	214	2	Dominican Peso
DZD	12	2	Algerian Dinar
EGP	818	2	Egyptian Pound
ERN	232	2	Eritrean Nakfa
ETB	230	2	Ethiopian Birr
EUR	978	2	Euro
FJD	242	2	Fijian Dollar
FKP	238	2	Falkland Islands Pound
GBP	826	2	Pound Sterling
GEL	981	2	Georgian Lari
GHS	936	2	Ghanaian Cedi
GIP	292	2	Gibraltar Pound
GMD	270	2	Gambian Dalasi
GNF	324	0	Guinean Franc
GTQ	320	2	Guatemalan Quetzal
GYD	328	2	Guyanese Dollar
HKD	344	2	Hong Kong Dollar
HNL	340	2	Honduran Lempira
HRK	191	2	Croatian Kuna
HTG	332	2	Haitian Gourde
HUF	348	2	Hungarian Forint
IDR	360	2	Indonesian Rupiah
ILS	376	2	Israeli New Shekel
INR	356	2	Indian Rupee
IQD	368	3	Iraqi Dinar
IRR	364	2	Iranian Rial
ISK	352	0	Icelandic Krona

Code	Num	Min	Name
JMD	388	2	Jamaican Dollar
JOD	400	3	Jordanian Dinar
JPY	392	0	Japanese Yen
KES	404	2	Kenyan Shilling
KGS	417	2	Kyrgyzstani Som
KHR	116	2	Cambodian Riel
KMF	174	0	Comorian Franc
KPW	408	2	North Korean Won
KRW	410	0	South Korean Won
KWD	414	3	Kuwaiti Dinar
KYD	136	2	Cayman Islands Dollar
KZT	398	2	Kazakhstani Tenge
LAK	418	2	Laotian Kip
LBP	422	2	Lebanese Pound
LKR	144	2	Sri Lankan Rupee
LRD	430	2	Liberian Dollar
LSL	426	2	Lesotho Loti
LYD	434	3	Libyan Dinar
MAD	504	2	Moroccan Dirham
MDL	498	2	Moldovan Leu
MGA	969	2	Malagasy Ariary
MKD	807	2	Macedonian Denar
MMK	104	2	Myanmar Kyat
MNT	496	2	Mongolian Togrog
MOP	446	2	Macanese Pataca
MRU	929	2	Mauritanian Ouguiya
MUR	480	2	Mauritian Rupee
MVR	462	2	Maldivian Rufiyaa
MWK	454	2	Malawian Kwacha
MXN	484	2	Mexican Peso
MYR	458	2	Malaysian Ringgit
MZN	943	2	Mozambican Metical

Code	Num	Min	Name
NAD	516	2	Namibian Dollar
NGN	566	2	Nigerian Naira
NIO	558	2	Nicaraguan Cordoba
NOK	578	2	Norwegian Krone
NPR	524	2	Nepalese Rupee
NZD	554	2	New Zealand Dollar
OMR	512	3	Omani Rial
PAB	590	2	Panamanian Balboa
PEN	604	2	Peruvian Sol
PGK	598	2	Papua New Guinean Kina
PHP	608	2	Philippine Peso
PKR	586	2	Pakistani Rupee
PLN	985	2	Polish Zloty
PYG	600	0	Paraguayan Guarani
QAR	634	2	Qatari Riyal
RON	946	2	Romanian Leu
RSD	941	2	Serbian Dinar
RUB	643	2	Russian Ruble
RWF	646	0	Rwandan Franc
SAR	682	2	Saudi Riyal
SBD	90	2	Solomon Islands Dollar
SCR	690	2	Seychellois Rupee
SDG	938	2	Sudanese Pound
SEK	752	2	Swedish Krona
SGD	702	2	Singapore Dollar
SHP	654	2	Saint Helena Pound
SLE	925	2	Sierra Leonean Leone
SLL	694	2	Sierra Leonean Leone (old)
SOS	706	2	Somali Shilling
SSP	728	2	South Sudanese Pound
SRD	968	2	Surinamese Dollar
STN	930	2	Sao Tome and Principe Dobra

Code	Num	Min	Name
SVC	222	2	Salvadoran Colon
SYP	760	2	Syrian Pound
SZL	748	2	Swazi Lilangeni
THB	764	2	Thai Baht
TJS	972	2	Tajikistani Somoni
TMT	934	2	Turkmenistan Manat
TND	788	3	Tunisian Dinar
TOP	776	2	Tongan Pa'anga
TRY	949	2	Turkish Lira
TTD	780	2	Trinidad and Tobago Dollar
TWD	901	2	New Taiwan Dollar
TZS	834	2	Tanzanian Shilling
UAH	980	2	Ukrainian Hryvnia
UGX	800	0	Ugandan Shilling
USD	840	2	US Dollar
UYU	858	2	Uruguayan Peso
UZS	860	2	Uzbekistani Som
VES	928	2	Venezuelan Bolivar Soberano
VND	704	0	Vietnamese Dong
VUV	548	0	Vanuatu Vatu
WST	882	2	Samoa Tala
XAF	950	0	CFA Franc BEAC
XAG	961	0	Silver
XAU	959	0	Gold
XCD	951	2	East Caribbean Dollar
XDR	960	0	Special Drawing Rights
XOF	952	0	CFA Franc BCEAO
XPB	964	0	Palladium
XPF	953	0	CFP Franc
XPT	962	0	Platinum
XTS	963	0	Test (ISO 4217 reserved)
YER	886	2	Yemeni Rial

Code	Num	Min	Name
ZAR	710	2	South African Rand
ZMW	967	2	Zambian Kwacha
ZWL	932	2	Zimbabwean Dollar

CLF is the only currency with 4 minor units. XTS is the ISO 4217 testing code; present in the table but should not appear in production data.

5.3 Cryptocurrencies

50 codes occupying `value_base_unit_t` slots **298 ... 347** (`BVN_CURRENCY_CRYPT0_FIRST ... BVN_CURRENCY_CRYPT0_LAST`). Like the fiat codes they have no named `bu *` enumerators — resolved by `bvn_parse_currency_str`, queried with `bvn_unit_is_crypto` / `bvn_currency_info`. `numeric_code = 0` for all.

Min = `minor_unit` = on-chain decimal places. E.g. `<uint:64,$BTC>` stores satoshis; divide by 10^8 to obtain BTC.

Code	Min	Subunit	Name
BTC	8	satoshi	Bitcoin
ETH	18	wei	Ethereum
SOL	9	lamport	Solana
XRP	6	drop	XRP
BNB	18	—	BNB
ADA	6	lovelace	Cardano
LTC	8	—	Litecoin
DOT	10	planck	Polkadot
XMR	12	piconero	Monero
ETC	18	—	Ethereum Classic
BCH	8	—	Bitcoin Cash
XLM	7	stroop	Stellar
FIL	18	—	Filecoin
ICP	8	—	Internet Computer
TRX	6	—	TRON
EOS	4	—	EOS
VET	18	—	VeChain
NEO	8	—	Neo

Code	Min	Subunit	Name
ZEC	8	—	Zcash
UNI	18	—	Uniswap
ARB	18	—	Arbitrum
SUI	9	—	Sui
TON	9	—	Toncoin
INJ	18	—	Injective
SEI	6	—	Sei
APT	8	—	Aptos
TAO	9	—	Bittensor
WIF	6	—	dogwifhat
DOGE	8	koinu	Dogecoin
LINK	18	—	Chainlink
USDT	6	—	Tether
USDC	6	—	USD Coin
AVAX	18	—	Avalanche
ATOM	6	—	Cosmos
POL	18	—	Polygon
NEAR	24	—	NEAR Protocol
ALGO	6	—	Algorand
HBAR	8	—	Hedera
AAVE	18	—	Aave
MKR	18	—	Maker
DAI	18	—	Dai
STX	6	—	Stacks
GRT	18	—	The Graph
LDO	18	—	Lido DAO
BONK	5	—	Bonk
PEPE	18	—	Pepe
SHIB	18	—	Shiba Inu
JUP	6	—	Jupiter
PYTH	6	—	Pyth Network
RUNE	8	—	THORChain

5.4 Currency Prefix Rules

All 24 SI prefixes are allowed on all currency units. IEC binary prefixes are forbidden on all currencies (`error_unit_illegal`).

Example	Meaning
<code>k~USD</code>	thousands of USD ($\times 10^3$)
<code>M~EUR</code>	millions of EUR ($\times 10^6$)
<code>G~ETH</code>	giga-ETH = Gwei scale ($\times 10^9$)
<code>m~USD</code>	milli-USD = one tenth of a cent ($\times 10^{-3}$)

6. Symbol Disambiguation

As of spec 1.0 a currency is written **only** with the `$` sigil (`$5.1`), so the bare form is always a physical-unit lookup and the namespaces never collide:

Token	Bare form (no <code>\$</code>)	<code>\$</code> -sigil form
<code>cup</code> / <code>CUP</code>	<code>cup</code> → US cup (<code>bu_cup</code>); <code>CUP</code> → <code>error_unit_illegal</code>	<code>\$CUP</code> → Cuban Peso (ISO 4217:192)
<code>BTU</code>	<code>BTU</code> → BTU (<code>bu_btu</code>); <code>Btu</code> , <code>btu</code> also accepted	(not ISO 4217)
<code>SOL</code>	<code>SOL</code> → <code>error_unit_illegal</code>	<code>\$SOL</code> → Solana (crypto)
<code>BAR</code>	<code>BAR</code> → <code>error_unit_illegal</code> ; use lowercase <code>bar</code>	(not ISO 4217)
<code>ERG</code>	<code>ERG</code> → <code>error_unit_illegal</code> ; use lowercase <code>erg</code>	(not ISO 4217)
<code>CAD</code> , <code>XAU</code>	<code>error_unit_illegal</code> (no physical unit)	<code>\$CAD</code> → Canadian Dollar; <code>\$XAU</code> → Gold (X-code)

No bare token is simultaneously a valid physical unit and a currency: currencies live entirely under `$` , physical units entirely without it.

Physical unit enum range: 1-133, 348-367, 368-371, and 372-377 (163 total) · Fiat: 134-297 (164) · Crypto: 298-347 (50) `BNV_VALUE_BASE_UNIT_COUNT` = 378 (`bu_ppb + 1`)